

Data 110-Fall 2026

Week 2

2 Visualizing data:

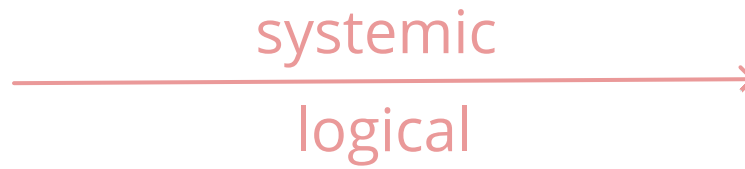
Mapping data onto aesthetics

Definition: Data visualization transforms data values into visual elements using a systematic and logical process.

Despite their differences, scatter plots, pie charts, and heatmaps share a common language in how data is represented.

- All visualizations map data values to quantifiable features in the final graphic.
- These features are called aesthetics (e.g., color, size, position).

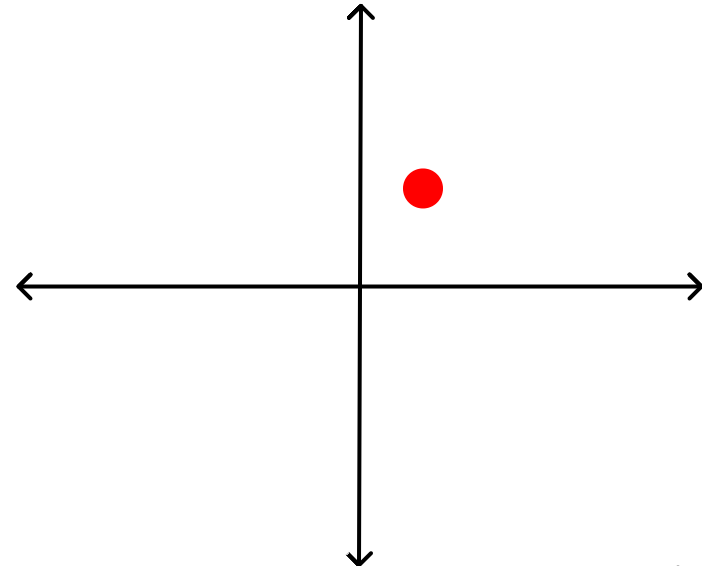
Data



visual

Example

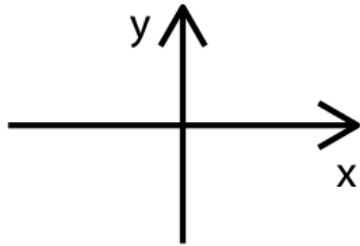
$(2, 3)$



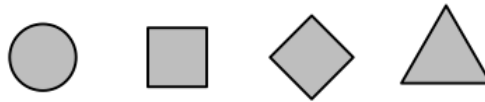
Key Aesthetics in Graphical Elements

- **Position:** Defined by x and y values in 2D graphics; other coordinate systems and dimensions are possible.
- **Shape, Size, Color:** Essential attributes for visibility, even in black-and-white visuals.
- **Line Attributes:** Width and dash-dot patterns for lines.
- **Text Attributes:** Font family, face, and size for text elements.
- **Transparency:** Used when graphical objects overlap.
- **Other Aesthetics:** Numerous additional attributes may be applied in data visualizations.

position



shape



size



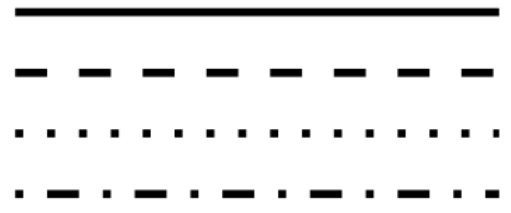
color

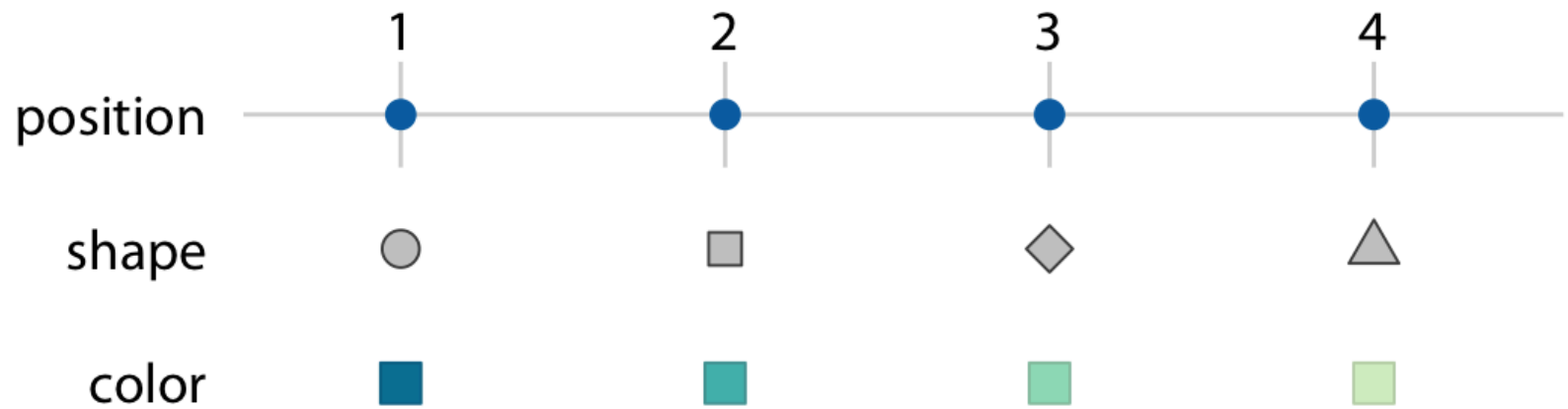


line width



line type





Aesthetics and Data Types

Two Groups of Aesthetics:

2. **Continuous Data:** Can represent values with arbitrarily fine intermediates (e.g., time duration, position, size, color, line width).
3. **Discrete Data:** Cannot represent intermediates (e.g., number of persons, shape, line type).

Table 2.1: Types of variables encountered in typical data visualization scenarios.

Type of variable	Examples	Appropriate scale	Description
quantitative/numerical continuous	1.3, 5.7, 83, 1.5×10^{-2}	continuous	Arbitrary numerical values. These can be integers, rational numbers, or real numbers.
quantitative/numerical discrete	1, 2, 3, 4	discrete	Numbers in discrete units. These are most commonly but not necessarily integers. For example, the numbers 0.5, 1.0, 1.5 could also be treated as discrete if intermediate values cannot exist in the given dataset.
qualitative/categorical unordered	dog, cat, fish	discrete	Categories without order. These are discrete and unique categories that have no inherent order. These variables are also called <i>factors</i> .
qualitative/categorical ordered	good, fair, poor	discrete	Categories with order. These are discrete and unique categories with an order. For example, “fair” always lies between “good” and “poor”. These variables are also called <i>ordered factors</i> .
date or time	Jan. 5 2018, 8:03am	continuous or discrete	Specific days and/or times. Also generic dates, such as July 4 or Dec. 25 (without year).
text	The quick brown fox jumps over the lazy dog.	none, or discrete	Free-form text. Can be treated as categorical if needed.

Definition:

An **ordered factor** is a categorical variable where the levels have a meaningful order or ranking. The sequence matters, and comparisons like "greater than" or "less than" make sense.

Example:

- **Month:** January < February < March < ... < December (logical order).
- **Education Levels:** Elementary < Middle School < High School < College.
- **Customer Satisfaction:** "Very Dissatisfied" < "Dissatisfied" < "Neutral" < "Satisfied" < "Very Satisfied."

Table 2.2: First 12 rows of a dataset listing daily temperature normals for four weather stations. Data source: NOAA.

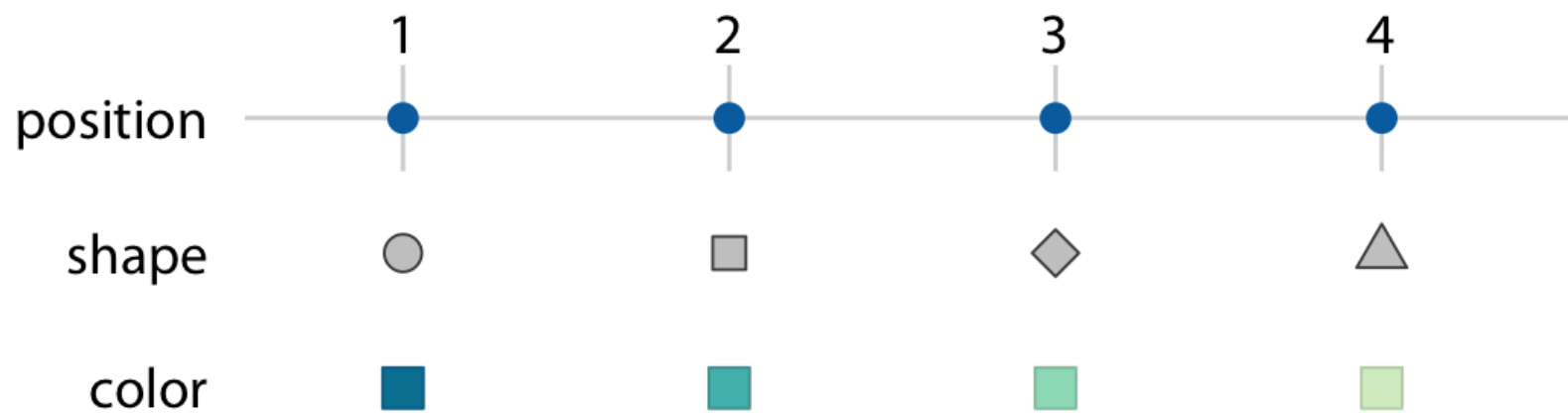
Month	Day	Location	Station ID	Temperature
Jan	1	Chicago	USW00014819	25.6
Jan	1	San Diego	USW00093107	55.2
Jan	1	Houston	USW00012918	53.9
Jan	1	Death Valley	USC00042319	51.0
Jan	2	Chicago	USW00014819	25.5
Jan	2	San Diego	USW00093107	55.3
Jan	2	Houston	USW00012918	53.8
Jan	2	Death Valley	USC00042319	51.2
Jan	3	Chicago	USW00014819	25.3
Jan	3	San Diego	USW00093107	55.3
Jan	3	Death Valley	USC00042319	51.3
Jan	3	Houston	USW00012918	53.8

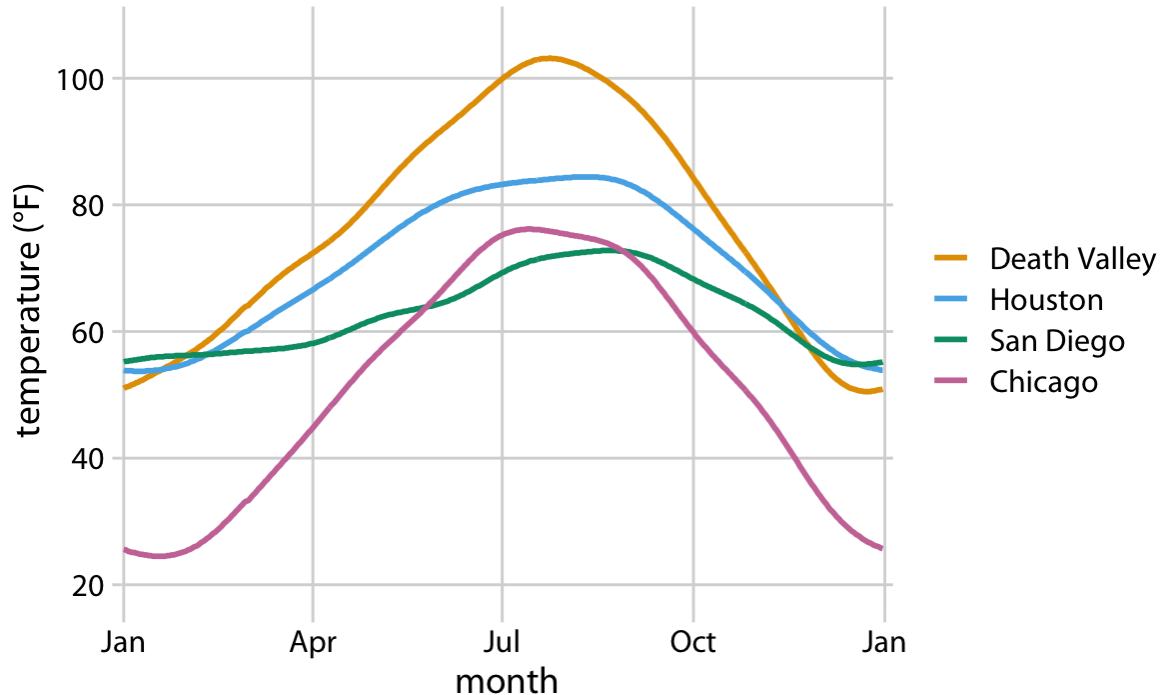
The dataset contains **five variables**:

- **Month** – Ordered factor
- **Day** – Discrete numerical value
- **Location** – Unordered factor
- **Station ID** – Unordered factor
- **Temperature** – Continuous numerical value

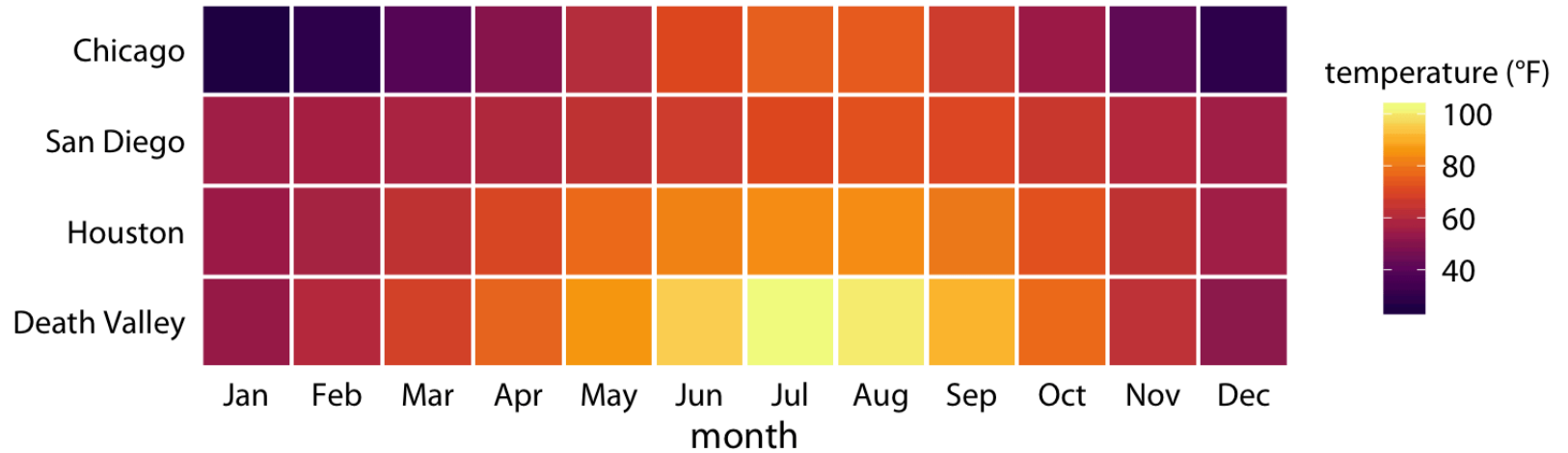
Scales map data values onto aesthetics

- **Mapping to Aesthetics:** Specifies how data values correspond to aesthetics (e.g., x-axis position, shape, color).
- **Scales:** Define a unique relationship between data and aesthetic values.
- **One-to-One Mapping:** Each data value must have exactly one aesthetic value and vice versa.
- **Importance:** If the mapping isn't one-to-one, the visualization becomes ambiguous.





- **Putting Theory into Practice:** The dataset is visualized using a line plot.
- **Mapping:**
 - **Temperature** → y-axis
 - **Day of the year** → x-axis
 - **Location** → Line color
- **Result:** A standard line plot showing temperature normals across four U.S. locations throughout the year.



- **Visualization Change:**
- - Temperature → **Color**
 - Month → **x-axis** (ordered factor)
 - Location → **y-axis** (unordered factor)
- **Representation:**
 - Uses **squares** instead of lines to better convey temperature differences through color.
- **Position Scales:**
 - Both **month** and **location** are **discrete** (not continuous).
 - **Month** is ordered, so it follows a natural sequence.
 - **Location** is unordered, so its order is flexible.
- **Chosen Order:** Locations are ordered **coldest (Chicago)** → **hottest (Death Valley)** for a better visual effect, but any order would be valid.

From

Fundamentals of Data Visualization *Claus O. Wilke*

